Chapter 11. The Seifert–van Kampen Theorem

Free products and free groups, the Seifert–van Kampen theorem, wedges of circles, attaching 2-cells

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The Seifert–van Kampen theorem computes the fundamental group of a space built from pieces whose groups are known. To state it we need some group theory.

11.1 Direct sums, free products and free groups

For abelian groups G_α, the **direct sum** $\bigoplus G_\alpha$ is characterized by the extension property: any family of homomorphisms $G_\alpha \to H$ (H abelian) extends uniquely to $\bigoplus G_\alpha \to H$. A **free abelian group** on $\{a_\alpha\}$ is $\bigoplus \mathbb{Z}a_\alpha$; its rank is well defined (rank of $G/2G$ as a $\mathbb{Z}/2$-vector space).

A group G generated by subgroups G_α with $G_\alpha \cap G_\beta = \{1\}$ is the **free product** $*G_\alpha$ if every element has a unique representation as a *reduced word* $g_1 \cdots g_n$, $g_i \neq 1$, consecutive letters from different G_α. Free products exist for any family (construct reduced words and let them act on the set of reduced words), and are characterized by the extension property: homomorphisms $G_\alpha \to H$ extend uniquely to $*G_\alpha \to H$ for every group H.

Theorem 11.1

If $G = G_1 * G_2$ and N_i is normal in G_i, then $G/\langle\langle N_1, N_2\rangle\rangle \cong (G_1/N_1) * (G_2/N_2)$, where $\langle\langle\cdot\rangle\rangle$ denotes the least normal subgroup containing the sets.

The **free group** on $\{a_\alpha\}$ is $*\langle a_\alpha\rangle$ with each $\langle a_\alpha\rangle \cong \mathbb{Z}$. Its abelianization $G/[G,G]$ is free abelian of the same rank, so free groups of different finite ranks are not isomorphic. A **presentation** $\langle a_\alpha \mid r_\beta\rangle$ is the quotient of the free group by the normal subgroup generated by the relators r_β.

11.2 The Seifert–van Kampen theorem

Theorem 11.2 (Seifert–van Kampen)

Let $X = U \cup V$ with U, V open and U, V, $U \cap V$ path connected, $x_0 \in U \cap V$. Let $\phi : \pi_1(U) * \pi_1(V) \to \pi_1(X)$ be induced by the inclusions. Then ϕ is surjective, and its kernel is the least normal subgroup containing all elements $i_1(g)^{-1} i_2(g)$, $g \in \pi_1(U \cap V, x_0)$, where i_1, i_2 are induced by $U \cap V \hookrightarrow U, V$. Equivalently, $\pi_1(X)$ is the pushout: homomorphisms $\pi_1(U) \to H$, $\pi_1(V) \to H$ agreeing on $\pi_1(U \cap V)$ combine uniquely to $\pi_1(X) \to H$.

Idea. Surjectivity is Theorem 9.11. For the kernel, given compatible ϕ_1, ϕ_2 define Φ on a loop by subdividing it into pieces lying in U or V, joining the break points to x_0, and multiplying the images. Independence of choices and of homotopy follows by subdividing a path homotopy $I \times I \to X$ into small squares lying in U or V and changing the factorization one square at a time. ∎

Corollary 11.3

If $U \cap V$ is simply connected, $\pi_1(X) \cong \pi_1(U) * \pi_1(V)$. If V is simply connected, $\pi_1(X) \cong \pi_1(U)/\langle\langle i_1 \pi_1(U \cap V)\rangle\rangle$.

11.3 The fundamental group of a wedge of circles

A **wedge of circles** is a Hausdorff space that is a union of subspaces $S_\alpha \cong S^1$ any two of which meet only in a common point p (with the coherent topology if there are infinitely many).

Theorem 11.4

The fundamental group of a wedge of circles S_α is free, with free generators represented by loops going once around each S_α.

Proof. For finitely many circles, induct with Corollary 11.3: thicken S_1 and $S_2 \vee \cdots \vee S_n$ by short arcs at p; their intersection is contractible. For infinitely many, use that a loop and a homotopy (compact) meet only finitely many circles, together with retractions onto finite subwedges. ■

So the figure eight has $\pi_1 \cong \mathbb{Z} * \mathbb{Z}$, which is non-abelian, and $\mathbb{R}^2$ minus n points (which deformation retracts to a wedge of n circles) has free fundamental group of rank n.

11.4 Adjoining a two-cell

Let $X = A \cup B$ with B a closed 2-cell attached along its boundary by $h : \partial B \to A$ (so X is A with a disc glued along the loop $f = h|_{S^1}$).

Theorem 11.5

The inclusion $A \hookrightarrow X$ induces a surjection $\pi_1(A, a) \to \pi_1(X, a)$ whose kernel is the least normal subgroup generated by the class of the attaching loop f (transported to a).

Proof. Apply Corollary 11.3 with V the open disc, $U = X$ minus the centre of the disc (which deformation retracts onto A), and $U \cap V$ an annulus with $\pi_1 \cong \mathbb{Z}$ generated by a loop parallel to f. ■

Attaching n-cells for $n \geq 3$ does not change π_1 (S^{n-1} is simply connected). Hence every group with a presentation is the fundamental group of a 2-dimensional complex: take a wedge of circles for the generators and attach a 2-cell for each relator.

11.5 The torus and the dunce cap

Example 11.1

(a) The torus is a wedge of two circles a, b with a disc attached along $aba^{-1}b^{-1}$: $\pi_1(T) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z} \times \mathbb{Z}$.
(b) The projective plane is a circle with a disc attached along a^2: $\pi_1(P^2) \cong \mathbb{Z}/2$, agreeing with Example 9.2.
(c) The n-fold **dunce cap** (a disc with its boundary wrapped n times around a circle) has $\pi_1 \cong \mathbb{Z}/n$. Every finite cyclic group, and indeed every finitely presented group, occurs as a fundamental group of a compact space.